

DS100 Final Project Report

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Dataset: Social Media and Mental Health

Introduction

In our Final Project, we chose to use the Social Media and Mental Health Dataset to demonstrate a well-rounded understanding of the population. We reviewed demographic factors, variables and their relationships with each other, and found correlations between digital use and lifestyle factors. We began by looking at the demographics of the sample, and seeing how mental health and digital habits vary. We continued by analyzing the lifestyle choices of the population and how that influences happiness and stress. Finally, we developed an understanding of how digital activity influences lifestyle and what that means for individuals online.

Data Cleaning/Pre-processing

Before conducting any analysis, we performed several pre-processing and data cleaning steps to ensure the dataset was suitable for visualization, hypothesis testing and linear regression modeling.

We firstly checked how many entries and columns we had. There are 500 samples and 10 key variables: **User_ID**, **Age**, **Gender**, **Daily_Screen_Time(hrs)**, **Sleep_Quality(1-10)**, **Stress_Level(1-10)**, **Days_Without_Social_Media**, **Exercise_Frequency(week)**, **Social_Media_Platform**, **Happiness_Index(1-10)**. These traits gave us summarization about each individual’s personal information, including lifestyle, digital habits and mental health condition.

Second, we examined whether there were missing values in the dataset. Fortunately, no missing values were found in the dataset. Each column has 500 entries with reasonable data types.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   User_ID                              500 non-null    object
1   Age                                  500 non-null    int64
2   Gender                              500 non-null    object
3   Daily_Screen_Time(hrs)              500 non-null    float64
4   Sleep_Quality(1-10)                 500 non-null    float64
5   Stress_Level(1-10)                  500 non-null    float64
6   Days_Without_Social_Media            500 non-null    float64
7   Exercise_Frequency(week)            500 non-null    float64
8   Social_Media_Platform               500 non-null    object
9   Happiness_Index(1-10)               500 non-null    float64
dtypes: float64(6), int64(1), object(3)
memory usage: 39.2+ KB
```

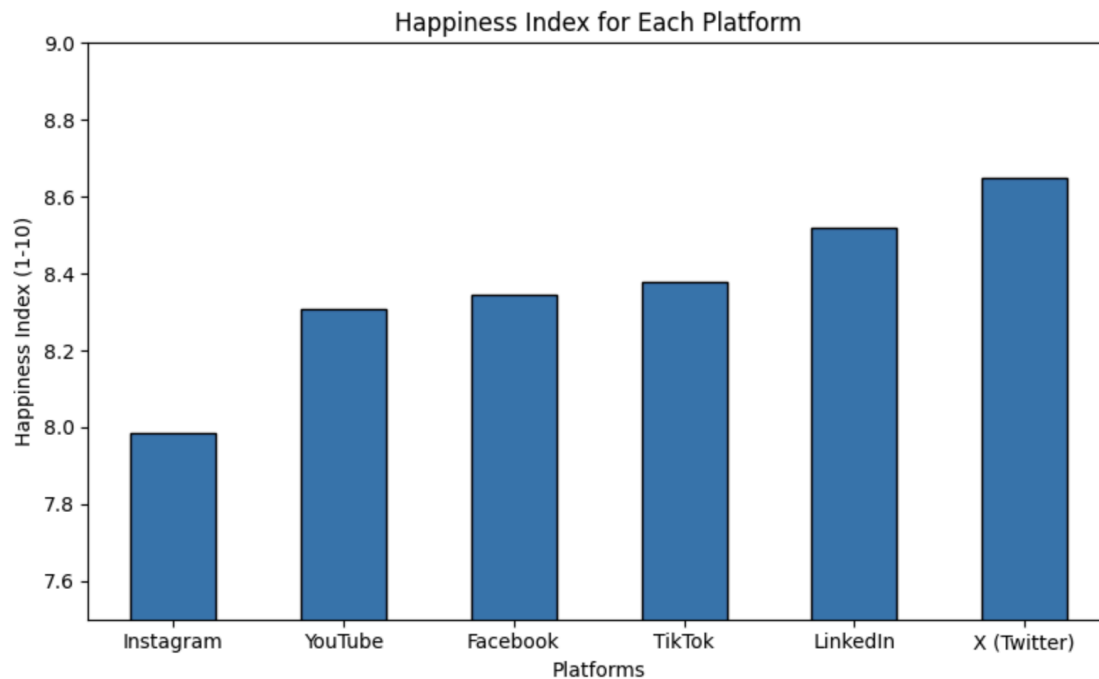
Finally, we checked that all variables fall within their expected range. For example, values in age, daily screen time, days without social media and exercise frequency were all non-negative; scale-based values in sleep quality, stress level and happiness index were not out of range.

	Age	Daily_Screen_Time(hrs)	Sleep_Quality(1-10)	Stress_Level(1-10)	Days_Without_Social_Media	Exercise_Frequency(week)	Happiness_Index(1-10)
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	32.988000	5.530000	6.304000	6.618000	3.134000	2.448000	8.370000
std	9.960637	1.734877	1.529792	1.542996	1.858751	1.428067	1.520000
min	16.000000	1.000000	2.000000	2.000000	0.000000	0.000000	4.000000
25%	24.000000	4.300000	5.000000	6.000000	2.000000	1.000000	7.000000
50%	34.000000	5.600000	6.000000	7.000000	3.000000	2.000000	9.000000
75%	41.000000	6.700000	7.000000	8.000000	5.000000	3.000000	10.000000
max	49.000000	10.800000	10.000000	10.000000	9.000000	7.000000	10.000000

Demographic and Social Media Platform Pattern Analysis (50 pts)

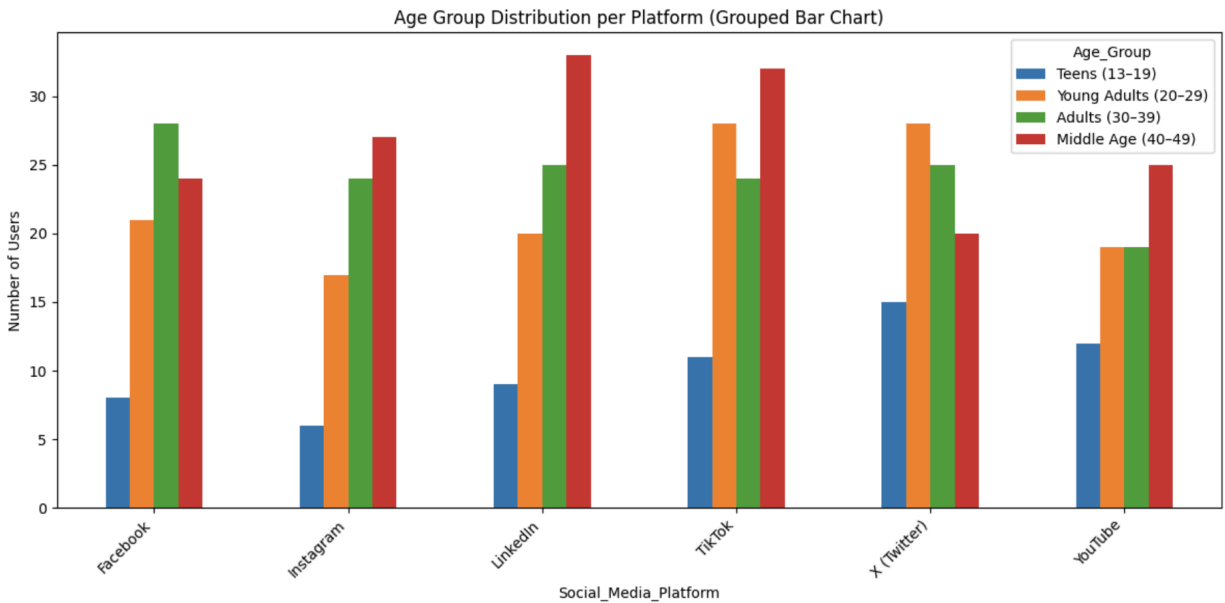
Research Question: How do mental health conditions and digital habits vary across different demographics (age & gender) and social media platforms?

 Visualization 1 – Average Happiness by Platform (5 pts)



A bar chart was chosen to compare the average happiness of each social media platform and determine which platforms are associated with the highest and lowest average happiness scores. By observing this plot, we could see that people who use X (Twitter) have the highest average happiness, while those who use Instagram have the lowest. It's easy for us to assert that the internet environment in Instagram causes low happiness. However, this graph does not show a causal relationship. Maybe happier individuals prefer certain platforms, or other factors cause this result.

 Visualization 2 – Age Distribution by Platform (5 pts)



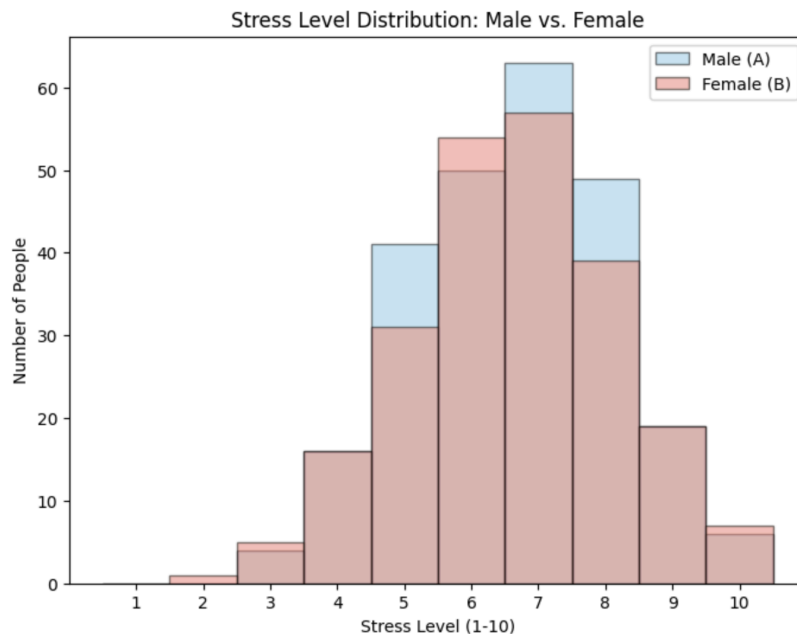
We used a grouped bar chart to explore the main age groups of users for each social media platform, hoping to make a decision of increasing the advertising of middle aged products on what social media platforms to maximize profits. The individuals in the dataset were grouped into 4 age groups: teenagers (13-19), young adults (20-29), adults (30-39) and middle age (40-49). From this plot, we found that Instagram, LinkedIn, TikTok and YouTube are dominated by middle age, Facebook is mainly used by Adults, and Young Adults are the main users on Twitter. In conclusion, if we want to sell more middle aged products, it's more beneficial to advertise on Instagram, LinkedIn, TikTok and YouTube, because more target groups will see the information about the products.

A/B Hypothesis Test – Gender and Stress Level (10 pts)

An A/B hypothesis test was chosen to investigate whether there is a significant difference in average **Stress_Level(1-10)** between males and females. The null and alternative hypothesis are:

- $H_0: \mu_{\text{male}} = \mu_{\text{female}}$ (the mean stress is the same for males and females).
- $H_1: \mu_{\text{male}} \neq \mu_{\text{female}}$ (the mean stress varies by gender).

We firstly created a single plot with two overlaid bar charts (one for male stress and one for female stress) to compare the Stress Level distributions. Because the average stress level for males and females overlap substantially, we guessed that we fail to reject the null hypothesis.



Our data fulfills the following three conditions:

1. Random: The data comes from a random sample.
2. Independence: The population is large enough to ensure that the sample size is less than 10% of the total population for both males and females.
3. Normality: The sample sizes for males and females are both greater than 30, so the Central Limit Theorem (CLT) assumes that both samples are normally distributed.

In this case, we could use the two-sample t-test.

T-Statistic: 0.21441355338573137
P-Value (Two-Tailed): 0.8303179322615026

The test yielded a t-statistic of 0.2144 and a p-value of 0.8303. Since the p value is greater than 0.05, we fail to reject the null hypothesis. There is not a statistical evidence that the population mean stress level of males is significantly different from that of females.

🔍 Bootstrap Hypothesis Test – Platform Type, Gender and Stress Level (10 pts)

A bootstrap hypothesis test is chosen to investigate whether the mean difference in **Stress_Level(1-10)** between gender minority users (other gender) and mainstream users (males and females) is larger on appearance /competition - focused social media platforms (TikTok, Instagram and LinkedIn) than on other platforms (YouTube, Twitter and Facebook). The null and alternative hypothesis are:

- $H_0: \mu_{LIT} = \mu_{other}$

The mean difference in stress level between gender minority users and mainstream users is the same for appearance/competition-focused platforms (TikTok, Instagram and LinkedIn) and other platforms (YouTube, Twitter and Facebook).

- $H_1: \mu_{LIT} > \mu_{other}$

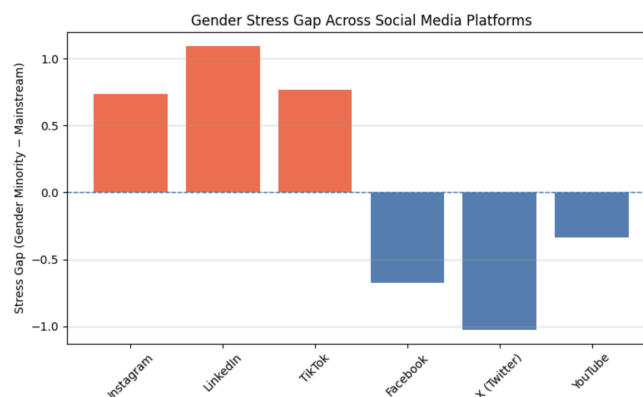
The mean difference in stress level between gender minority users and mainstream users is larger on appearance/competition-focused platforms (TikTok, Instagram and LinkedIn) than on other platforms (YouTube, Twitter and Facebook).

Number of Individuals in Mainstream Group: 477
Number of Individuals in Gender Minority: 23

We firstly count the number of individuals in the Mainstream Group and the Gender Minority. Because the sample size for the Gender Minority is less than 30, we cannot assure that this sample is normally distributed. Thus, we should use a bootstrap hypothesis test.

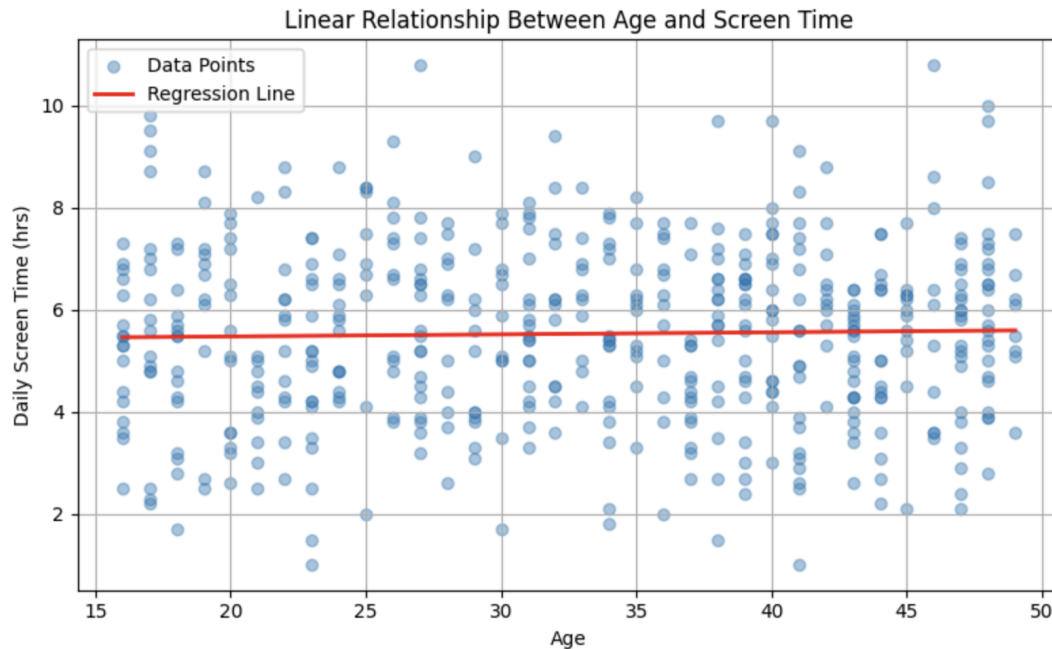
Observed difference in stress gap: 1.543
Bootstrap p-value: 0.0076

The test yielded a **p-value of 0.0076**. Since the p-value is far less than 0.05, we should reject the null hypothesis. There is a strong statistical evidence that the mean difference in stress level between gender minority users and mainstream users is larger on appearance/competition-focused platforms (TikTok, Instagram and LinkedIn) than on other platforms (YouTube, Twitter and Facebook).



Linear Regression – Predicting Daily Screen Time from Age (10 pts)

We want to use a linear regression model to explore whether we could predict an individual's daily screen time based on his or her age. We guessed that younger individuals, such as teens, have lower daily screen time, because some schools limit electronic devices usage. We modeled the relationship between Age and Daily_Screen_Time(hrs).



Intercept: 5.394950764249314
Slope: 0.004093889770543416
R square: 0.0005524702732515596

We got the plot and the values above.

- The slope of 0.0041 means that if an individual grows a year older, he or she will predictably spend more 0.0041 hours on digital screens per day.
- The intercept of 5.395 means that a newborn baby (0 year old) may spend 5.395 hours on screen. It's a non-sense intercept.
- R square of 0.00055 represents that there are only 0.055% of variances in daily screen time that can be explained by age.

Since the correlation coefficient (slope) is close to 0 and the regression line is almost horizontal, we cannot use age to predict an individual's daily screen time.

🎯 Advanced Analysis: ANOVA-- Average Daily Screen Time and Platforms (10 pts)

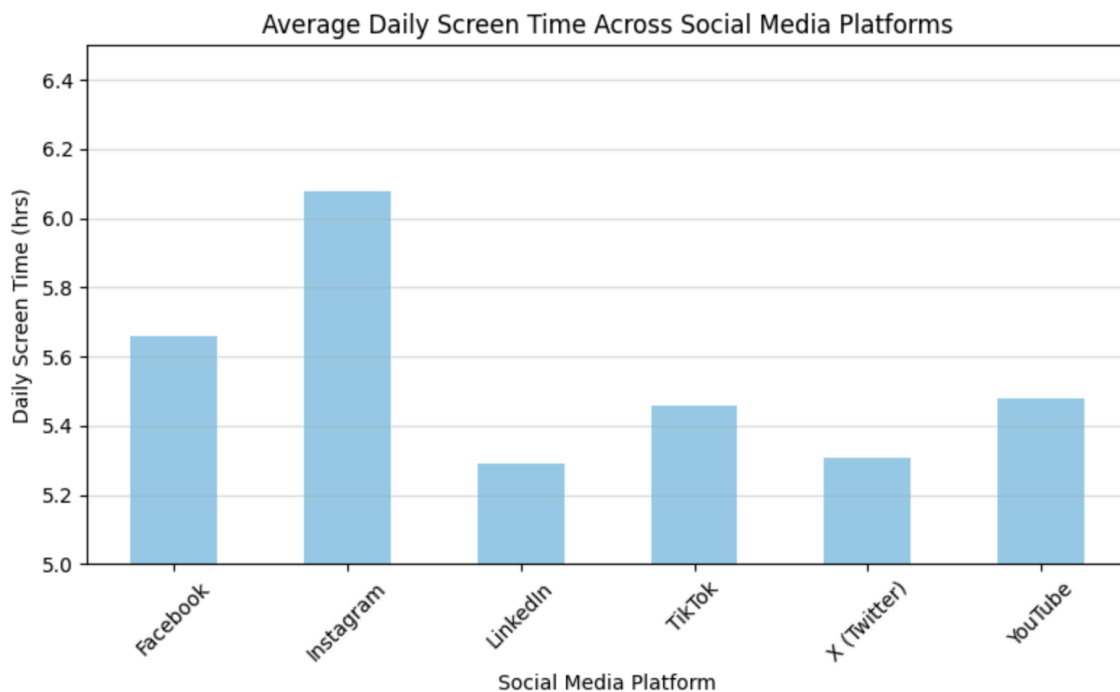
An ANOVA (Analysis of Variance) hypothesis test is chosen to investigate whether the mean **Daily_Screen_Time(hrs)** differ significantly across social media platforms. The null and alternative hypotheses are:

- $H_0: \mu_{\text{Facebook}} = \mu_{\text{Instagram}} = \mu_{\text{LinkedIn}} = \mu_{\text{TikTok}} = \mu_{\text{Twitter}} = \mu_{\text{YouTube}}$
The mean daily screen time of users is the same for 6 platforms).
- H_1 : At least one platform has a different mean daily screen time.

Considering that the amount of experimental variables is greater than two, we should use the ANOVA test.

ANOVA F-statistic: 2.267482397930763
p-value: 0.04676561087839459

The test yielded a **f-statistic of 2.2675** and a **p-value of 0.04676**. Since the p-value is less than 0.05, we should reject the null hypothesis. There is a strong statistical evidence that there's at least one social media platform in which the users spend more or less time on it.



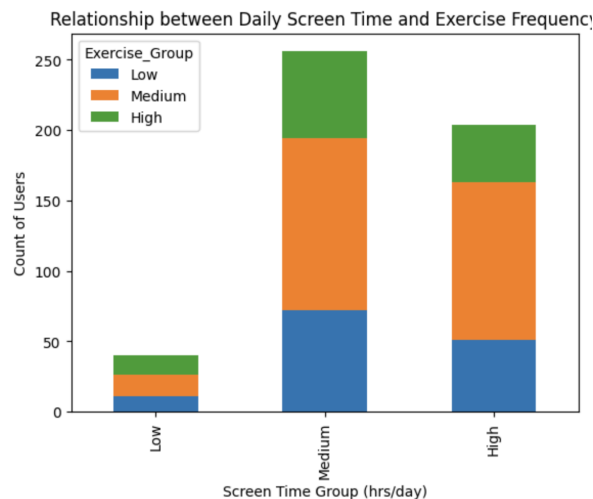
We are curious that people spend more or less time on which social media platforms. By plotting and observing, we drew a conclusion that people spend more time on Instagram.

Lifestyle and Mental Health Analysis

Research Question: How does lifestyle (screen time, exercise, sleep habits) influence mental health (stress level and happiness index)?

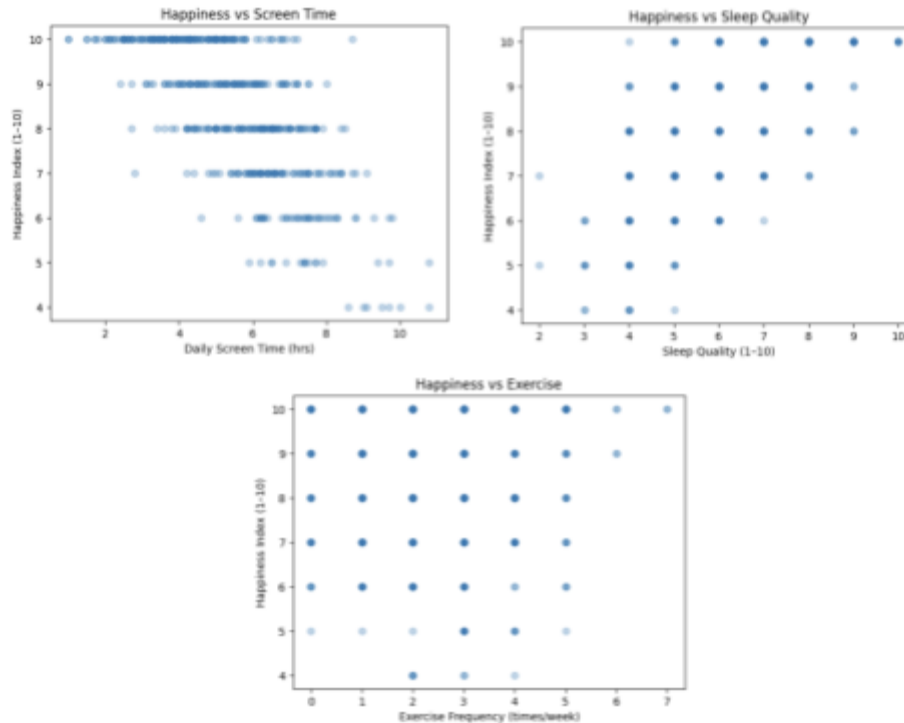
Visualization

First, a stacked bar chart was used to compare screen time and exercise frequency. 'Daily_Screen_Time(hrs)' and 'Exercise_Frequency(week)' were used to visualize. The exercise groups low, medium, and high were color coded and displayed across screen time groups low, medium, and high.



The visualization shows that most people fall into medium and high screen time. Within these groups, medium and low exercise frequency is more common. Although low screen time is less common, there is a greater proportion of medium and high exercise levels compared to medium and high screen time groups. This suggests a negative relationship between screen time and exercise frequency. Lower screen time is correlated to higher exercise frequency while high screen time is correlated to low or medium exercise frequency.

Second, scatter plots were used to compare happiness with screen time, exercise, and sleep quality. The variables 'Daily_Screen_Time(hrs)', 'Exercise_Frequency(week)', and 'Sleep_Quality(1-10)' were used along with 'Happiness_Index(1-10)'. The objective was to visualize correlation between lifestyle choices and happiness.



The visualization shows a negative correlation between happiness and screen time, a positive relationship between happiness and sleep quality, and no correlation between happiness and exercise. Therefore, higher screen time is associated with lower happiness, higher sleep quality is associated with higher happiness, and there is no significant relationship between happiness and exercise.

Hypothesis Test

First, hypothesis testing was used to see if a lifestyle with low screen time, high exercise frequency, and good sleep quality has a higher happiness index than another lifestyle with high screen time, low exercise, and poor sleep quality. The analysis used variables 'Daily_Screen_Time(hrs)', 'Exercise_Frequency(week)', and 'Sleep_Quality(1-10)' to represent lifestyle choices, and 'Happiness_Index(1-10)' to measure happiness. A two-sample t-test compares happiness to two contrasting lifestyles. This dataset acts as a sample of the broader population. The null hypothesis is that a lifestyle doesn't have an effect on happiness. The alternative hypothesis is that lifestyle does have an effect on happiness.

t-statistic: 10.24420683439111
 degrees of freedom: 29.721411538819122
 p-value (one-tailed): 1.4444001550373287e-11

The p-value is almost 0 which is far less than the chosen significance level of 0.05. Therefore, we can reject the null hypothesis, meaning there is statistically significant evidence that a lifestyle has an effect on happiness.

Second, hypothesis testing was used to determine if lifestyle choices correlate with stress levels. A/B testing was used to compare 'Stress_Level(1-10)' with a "healthy" lifestyle (low screen time, high exercise, high sleep quality) and an "unhealthy" lifestyle (high screen time, low exercise, lower sleep quality). The null hypothesis is that lifestyle doesn't have an effect on stress. The alternative hypothesis is that lifestyle does have an effect on stress.

t-statistic: -9.74784251545987
p-value: 4.002089624735236e-13

The p-value is far below the chosen significance level of 0.05. Therefore, we can reject the null hypothesis. This concludes that there is statistically significant evidence that individuals with a healthy lifestyle have lower stress levels than those with an unhealthy lifestyle.

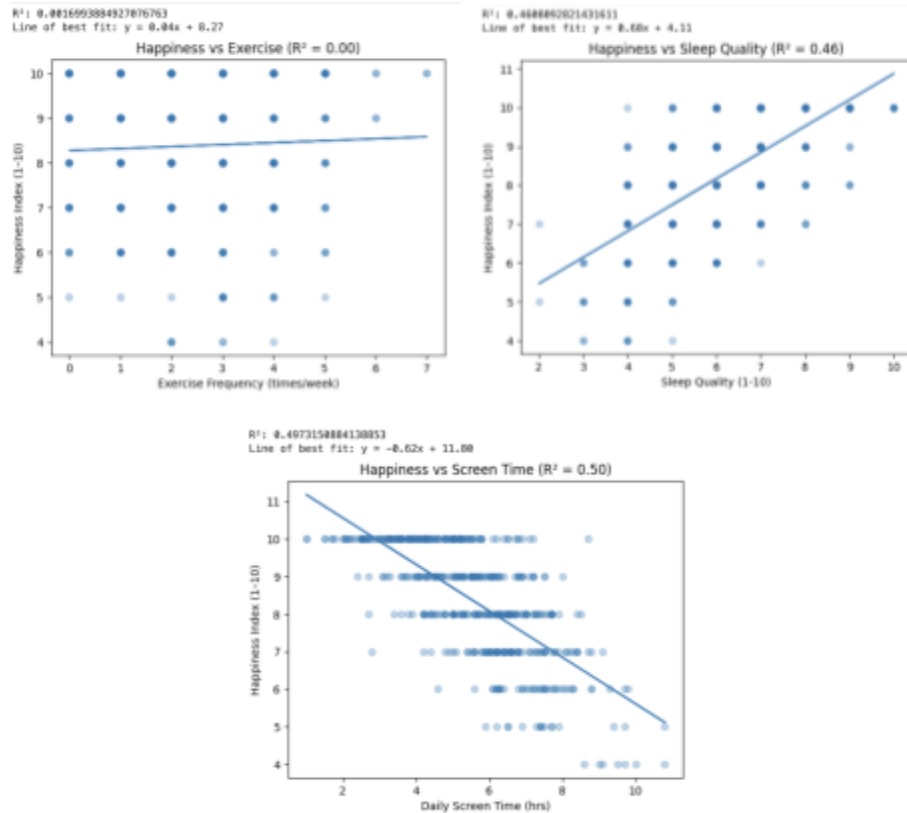
Linear Regression

First, we modeled the relationship between lifestyle ('Daily_Screen_Time(hrs)', 'Sleep_Quality(1-10)', 'Exercise_Frequency(week)') and 'Happiness_Index(1-10)'. We wanted to observe the extent that lifestyle is related to happiness.

Intercept: 8.47895866807058
Coefficients:
screen_time: -0.39577203115705933
exercise: -0.013486667097604793
sleep: 0.3360843948735538

The intercept is 8.479, meaning when screen time hours, exercise frequency, and sleep quality equal 0, the predicted happiness is 8.479 out of 10. It's essentially a baseline happiness level.

The coefficients represent the rate that each lifestyle choice affects happiness. For each additional hour of screen time, the happiness index decreases by 0.395 units. For every additional exercise happiness decreases by 0.013 units. For each unit increase in sleep quality, happiness increases by 0.336 points.



The R^2 value shows how much of the change in happiness can be explained by one of the lifestyle factors. Since we measured multiple relationships, there is an R^2 value for each lifestyle variable. The R^2 value for Happiness vs Exercise is 0.0017, meaning exercise has essentially no effect on happiness. The R^2 value for Happiness vs Sleep Quality is 0.4608, meaning sleep quality has a moderate-strong effect on happiness. The R^2 value for Happiness vs Screen Time is 0.4973, meaning screen time has a strong effect on happiness.

Again, because we modeled three different regression models, there are three individual line-of-best-fits for each lifestyle variable. The slope for the line of best fit for Happiness vs Exercise is 0.04, meaning there is not significant evidence to establish a conclusion. The slope for the line of best fit for Happiness vs Sleep Quality is 0.68, meaning there is a strong positive association between sleep quality and happiness. The slope for the line of best fit for Happiness vs Screen Time is -0.62, meaning there is a strong negative association between screen time and happiness.

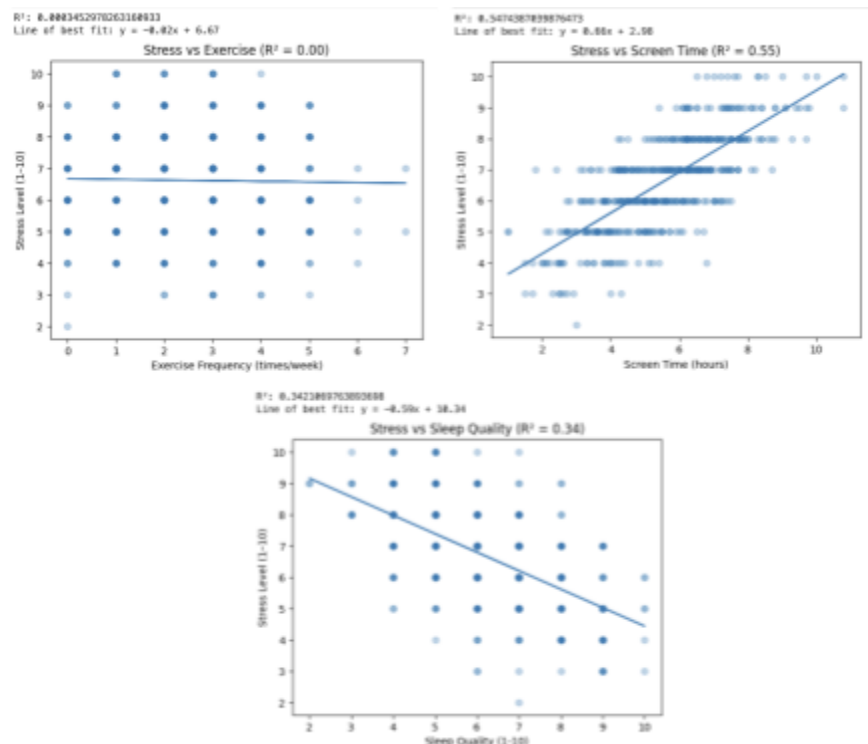
The model suggests that more screen time lowers happiness and better sleep increases happiness. The coefficient of exercise is so low that the model shows no significant relationship between exercise and happiness.

Second, we modeled the relationship between lifestyle ('Daily_Screen_Time(hrs)', 'Sleep_Quality(1-10)', 'Exercise_Frequency(week)') and 'Stress_Level(1-10)'. We wanted to observe the extent that lifestyle is related to stress.

Intercept: 3.3099484370414576
Coefficients:
screen_time: 0.6294990686938633
exercise: 0.057107624686828146
sleep: -0.04963162312053987

The intercept is 3.310, meaning when screen time hours, exercise frequency, and sleep quality equal 0, the predicted stress level is 3.310 out of 10. It's essentially a baseline stress level.

The coefficients represent the rate that each lifestyle choice affects stress. For each additional hour of screen time, the stress level increases by 0.629 units. For every additional exercise stress increases by 0.057 units. For each unit increase in sleep quality, stress decreases by 0.050 points.



The R^2 value shows how much of the change in stress can be explained by one of the lifestyle factors. Since we measured multiple relationships, there is an R^2 value for each lifestyle variable. The R^2 value for Stress vs Exercise is 0.0003, meaning exercise has essentially no effect on stress. The R^2 value for Stress vs Sleep Quality is 0.3421, meaning sleep quality has a moderate effect on stress. The R^2 value for Stress vs Screen Time is 0.5474, meaning screen time has a high effect on stress.

Again, because we modeled three different regression models, there are three individual line-of-best-fits for each lifestyle variable. The slope for the line of best fit for Stress vs Exercise is -0.02, meaning there is not significant evidence to establish a conclusion. The slope for the

line of best fit for Stress vs Sleep Quality is -0.59 , meaning there is a strong negative association between sleep quality and stress. The slope for the line of best fit for Stress vs Screen Time is 0.66 , meaning there is a strong positive association between screen time and stress.

The model suggests that more screen time increases stress levels, and higher sleep quality slightly decreases stress levels. The coefficient of exercise is so low that the model shows no significant relationship between exercise and stress.

Digital Behavior for Mental Health Prediction Analysis

Research Question:

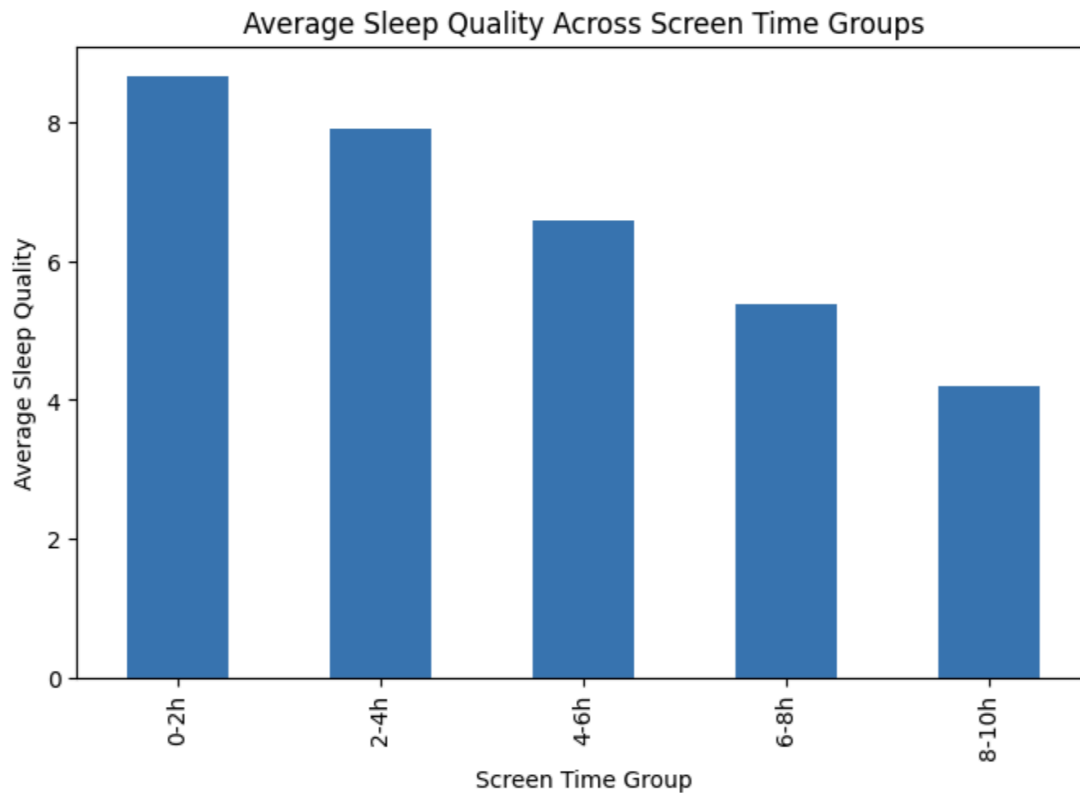
How does the frequency of digital activity, measured through screen time, influence lifestyle factors such as sleep quality and exercise frequency?

Part 1:

To form the foundation of our understanding of lifestyle factors such as sleep quality and exercise frequency in this data set, we explore these outcome variables via visualizations.

Visualization 1

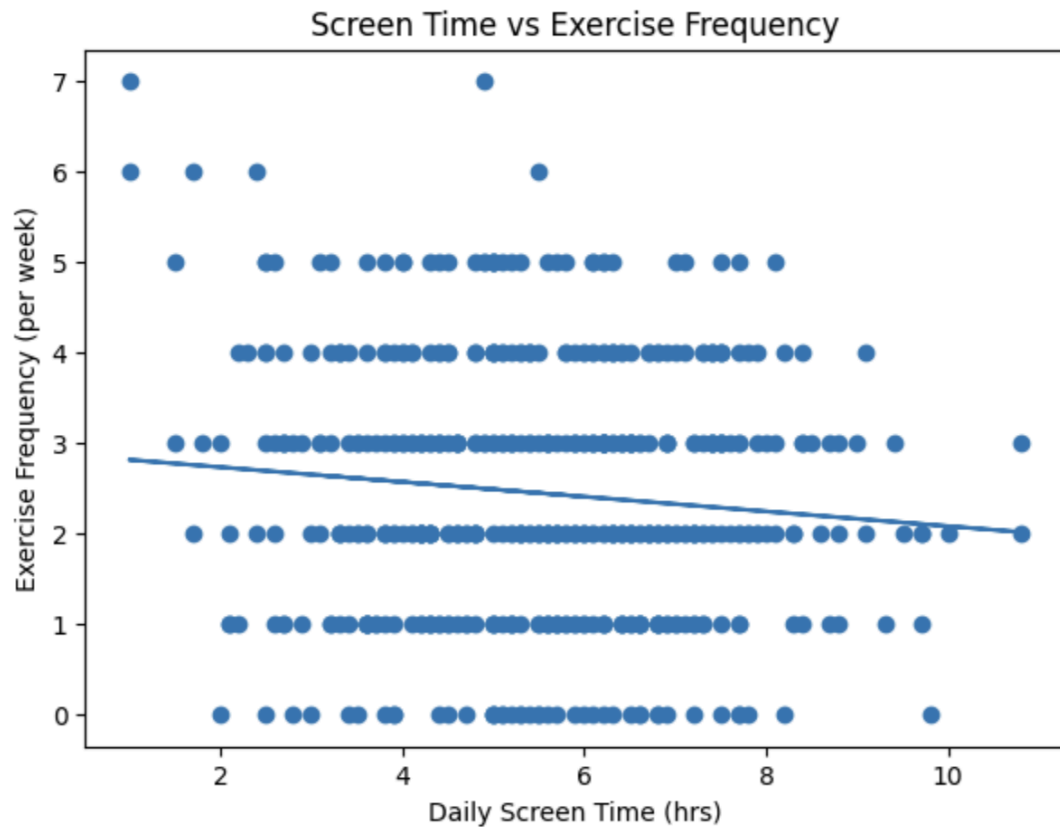
- Bar Chart of Screen Time vs Sleep Quality



- A Bar Chart was created to shape our understanding of the relationship between Screen Time Groups and Average Sleep Quality using the variables *screen_group* and *Sleep_Quality(1-10)*. As we can tell, there is a strong negative correlation between these two variables, concluding that the increase in screen time results in a decrease in sleep quality.

Visualization 2

- Scatter Plot of Screen Time vs Exercise Frequency



- A Scatter Plot was created to shape our understanding of the relationship between Screen Time Groups and Exercise Frequency using the variables *screen_group* and *Exercise_Frequency(week)*. As we can tell, there is a negative correlation between these two variables, even if it isn't as strong as the relationship above in the prior visualization. We can conclude that the increase in screen time results in a slight decrease in exercise frequency .

Part 2:

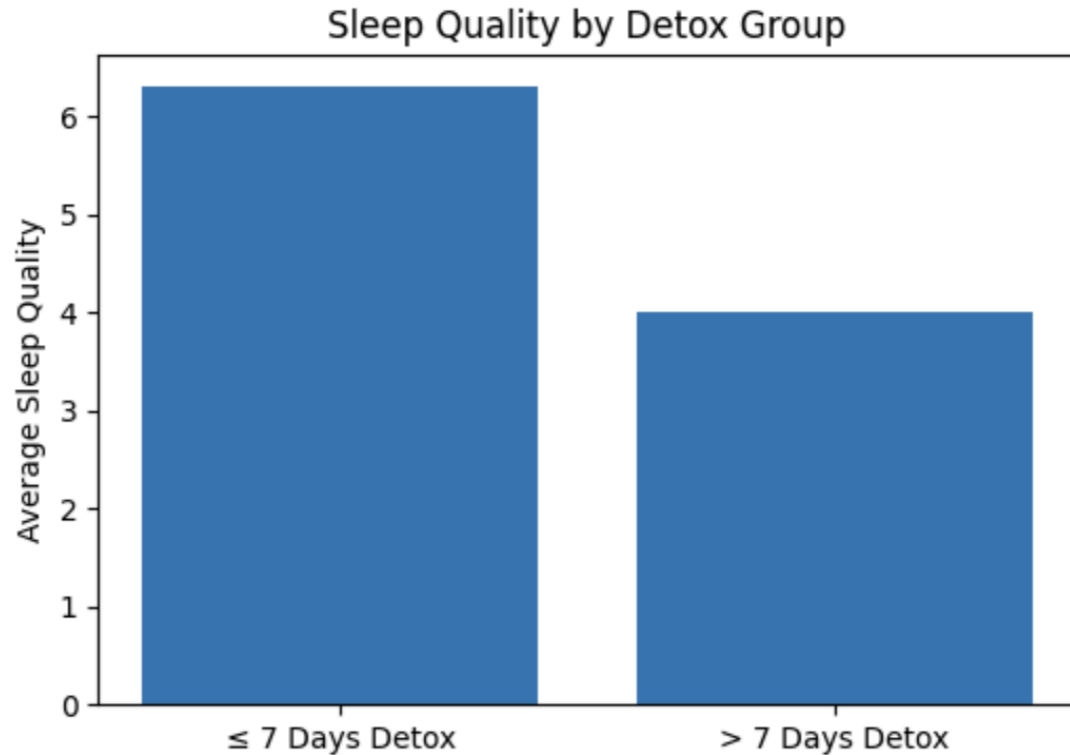
Furthering our familiarity of these variables, we can use A/B Hypothesis Testing with mental health and lifestyle outcome variables to compare two groups of students that differ in their digital behavior. For each A/B Test, we examine whether the standard time spent away from media (7 days) - using variable *screen_group* - could help us decide whether or not we can make future predictions about the population. Along with the bar graph, we are using a P value to test the significance of these findings; if the p-value is less than 0.05, we reject the null hypothesis and conclude that detox time is associated with a significant difference in the outcome, but if the p-value is greater than 0.05, we fail to reject the null hypothesis, meaning there is not enough evidence to say detox time affects the outcome. We also compute the T-value to see if there is even any clear difference between the 2 groups.

Hypothesis Test 1

- Detox Time and Sleep Quality

T-stat: 1.50952548072114

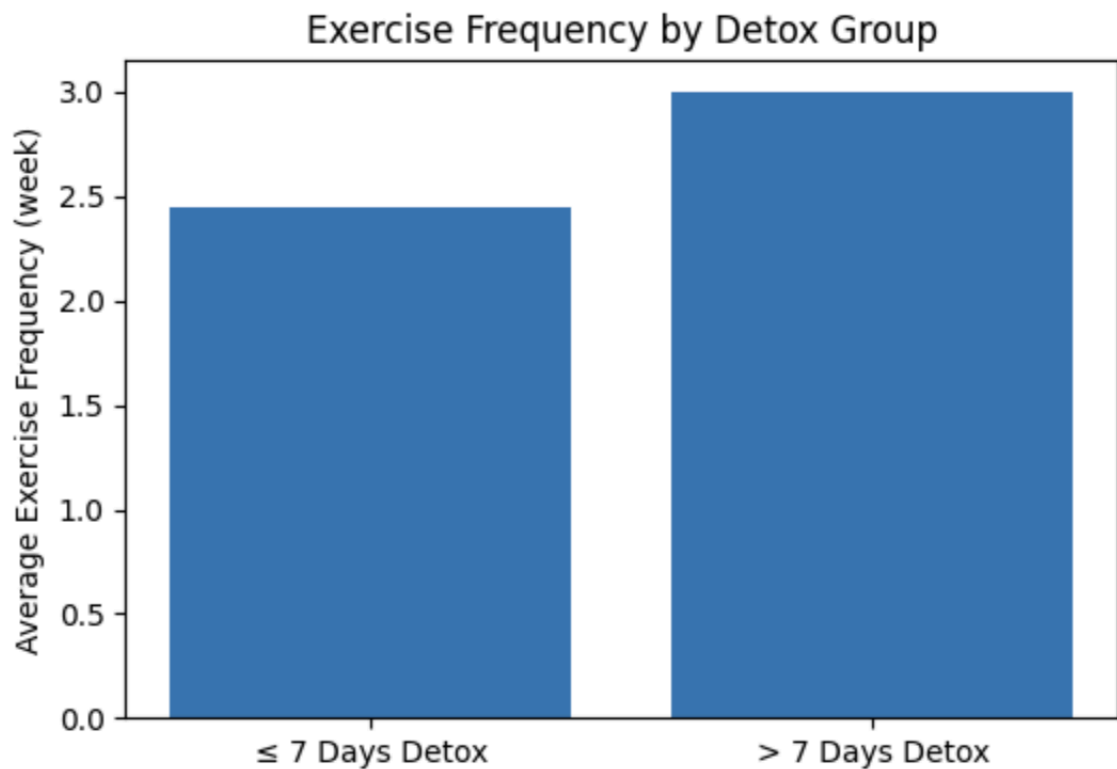
P-value: 0.13179872922034838



-
- As seen from this bar chart and the computed T-value, there is a clear difference between the averages of these two screen time groups. Although the results seem statistically significant, the P-value tells us otherwise, that although the outcome is informative, it's not enough to reject the null hypothesis.

Hypothesis Test 2

- Detox Time and Exercise Frequency
T-stat: -0.38659376463875933
P-value: 0.6992221027018033



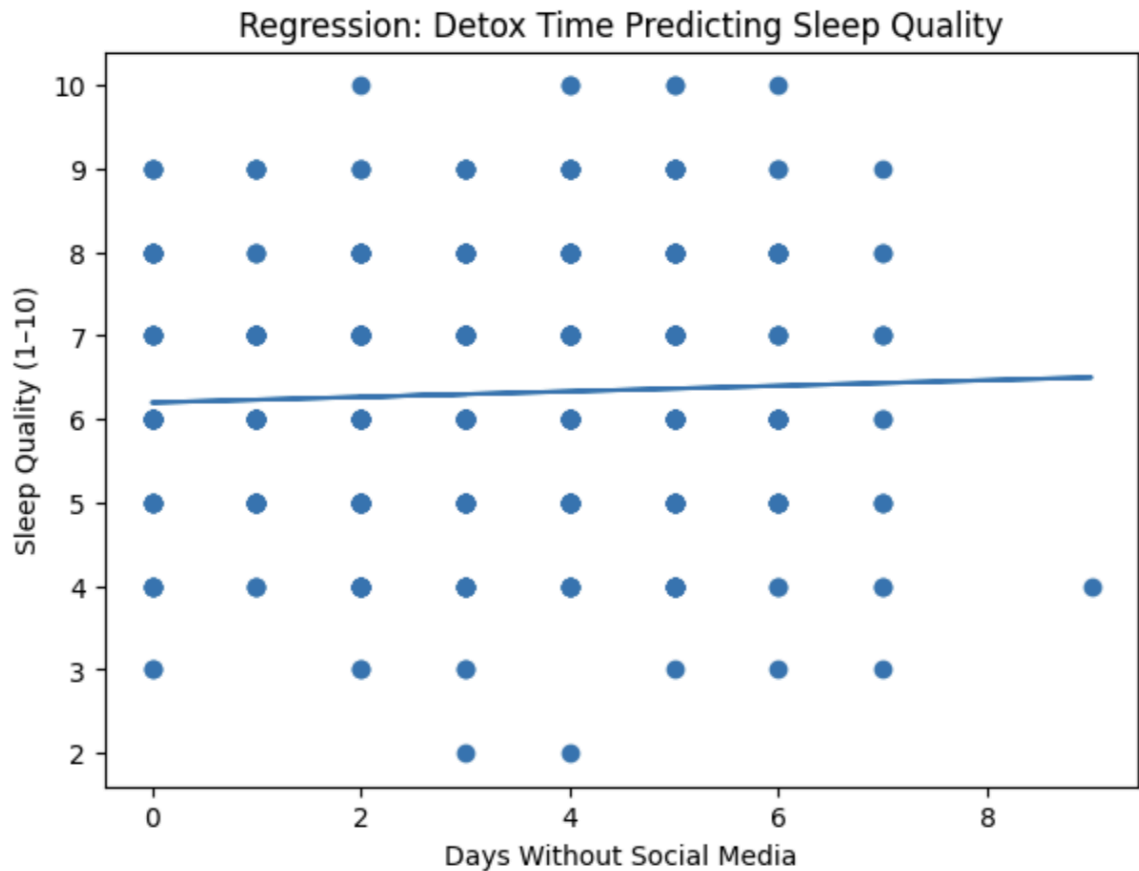
- As seen from this bar chart and the computed T-value, there is also a clear difference between the averages of these two screen time groups. Similar to the prior hypothesis test, the P-value tells us we do not have enough difference to reject the null hypothesis.

Part 3:

In the last section of lifestyle analysis, in correlation to *Days_Without_Social_Media*, we use Linear Regression to predict outcome variables *Sleep_Quality(1-10)* and *Exercise_Frequency(week)*. Rather than comparing group averages using A/B Testing as we did above, this part examines detox time as a continuous variable and evaluates how sleep quality changes as detox time increases/decreases.

Linear Regression 1

- Detox Time Predicting Sleep Quality



Slope: 0.03342880775303525

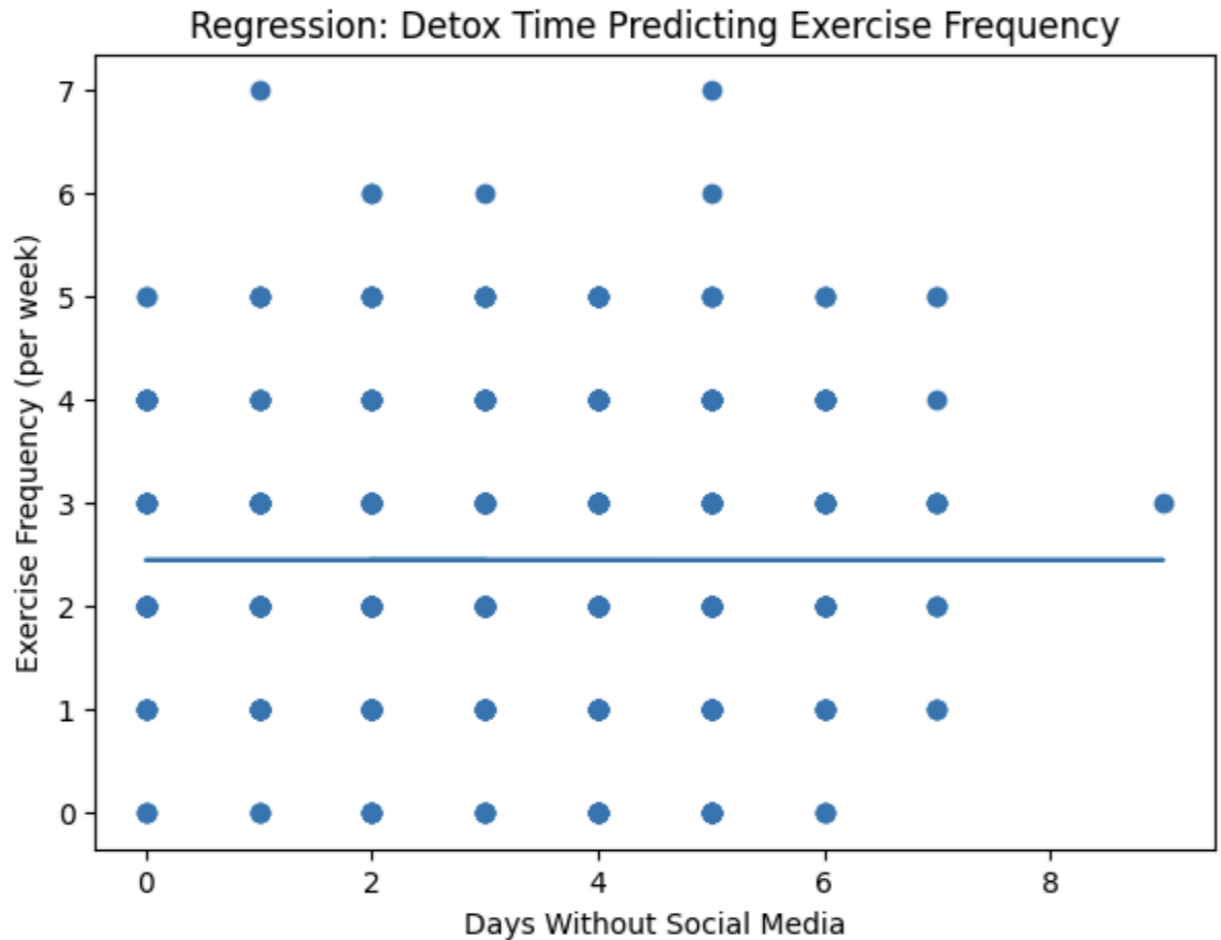
Intercept: 6.199234116501991

- R^2 : 0.001649753593467751

- This scatter plot with the line showing linear regression tells us that although there is a predicted relationship shown by the slope = 0.03, it is very minimal. In other words, over a 10 day span of no digital media use, this graph predicts that the sleep quality will increase by 0.03 points. We can determine that even though there is a positive relationship, it is very weak and not strongly predictive. The R^2 value of 0.00165 is small but still present, telling us that only about 0.17% of the variation in Sleep Quality can be explained by Days Without Social Media; an R^2 value of 1 would assume a perfect positive linear relationship between two variables.

Linear Regression 2

- Detox Time Predicting Exercise Frequency Exercise Frequency



Slope: $-9.280624028522064 \times 10^{-6}$

Intercept: 2.4480290854757065

- $R^2: 1.4591483576964492 \times 10^{-10}$
- This scatter plot with the line for linear regression seems to be almost flat, with the slope $= -0.000009$ meaning that it is almost effectively 0. This tells us there is no future for predicting exercise frequency directly correlated with days without social media. We can assume from this chart that exercise habits appear independent of detox behavior in this dataset. Additionally, the R^2 value being almost 0 is an indication that there is no meaningful linear relationship between social media detox time and exercise habits in this dataset.

Overall Insights and Conclusions

From the demographic and social media platform analysis, we found that average happiness and daily screen time vary across platforms, and that different platforms attract different age groups. To be specific, Instagram and Twitter are separately associated with lowest and highest happiness, and middle ages are the main users on LinkedIn, TikTok, YouTube and Instagram. With these conclusions, we can make financial decisions, such as advertising on certain platforms to attract more target groups. Additionally, A/B testing shows no statistical significance difference in average stress level between males and females. However, a bootstrap analysis reveals that gender minorities have a significant larger stress level than mainstream users in appearance- and competition- focused platforms (TikTok, LinkedIn, Instagram) than other platforms (YouTube, Twitter, Facebook). We should continuously strive for reducing discrimination and achieving equality.

The main conclusions taken from the lifestyle and mental health analysis come from the hypothesis testing and linear regression models. When looking at happiness, these tests show that happiness and sleep quality have a moderate-strong positive correlation, happiness and screentime have a strong negative correlation, and happiness and exercise have no significant correlation. When analyzing stress, stress and screen time have a strong positive correlation, stress and sleep quality have a moderate negative correlation, and stress and exercise have no significant correlation. This shows that ideal mental health (low stress and high happiness) comes from low screen time and high sleep quality. Exercise doesn't have a significant effect.

The key takeaways from Topic 3, Digital Behavior for Mental Health Prediction Analysis, state that grouping of Detox Time (Days Spent Without Social Media) mean averages portray visible differences in Exercise Frequency and Sleep Quality. While this is true, A/B Testing and Linear Regression, with the help of R^2 values, prove that these differences are still not statistically significant, so while we can try to use this data to make future predictions, there is no tangible evidence to follow our hypothesis.

AI Statement

For this project, AI was used to assist in writing code for visualization, statistical methods, and debugging. AI was used to edit the written explanations to ensure they fulfilled project requirements. All data analysis and conclusions were completed by the authors. AI was used as a learning aid rather than a substitute for original work.